

Olivia Fan

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EDUCATION

University of Toronto, St. George Campus

Toronto, ON

BASc Mechanical Engineering + PEY Co-op, Minor in Engineering Business

2024 – 2029

- Sessional GPA: 3.88; Cumulative GPA: 3.73; Dean's List all semesters to date
- **Relevant Courses:** Mechanics, Dynamics, Calculus I & II, Differential Equations, Chemistry and Material Science, Linear Algebra, Circuits, Fluid Mechanics, Control Systems

HIGHLIGHTS OF SKILLS AND ACHIEVEMENTS

CAD / Design: SolidWorks, Altium

Structural Analysis: ANSYS

Programming: Python, MATLAB, basic C with Arduino

Prototyping: Experience with 3D printing PLA, ABS and resin and operation/maintenance of 3D printers

Languages: Fluent in English, French, and Mandarin

Awards: Paul Cadario Social Impact Award, Dean's Merit Award, MIE Admission Scholarship

PROFESSIONAL EXPERIENCES

Surgical Robotics Researcher at PCIGITI

May 2026 – Aug. 2026

SickKids Hospital

Toronto, ON

- Developed and refined mechanical hardware components for a 3mm wrist and elbow transoral surgical tool (TORS), focusing on miniaturization, dexterity, and mechanical reliability
- Designed new tooltip joints with an analytical friction reduction of up to 400% in certain joints
- Captured and processed motor and joint motion data, conducting analytical calculations to evaluate system precision, trajectory tracking, and repeatability against theoretical models
- Collaborated with surgeons, biomedical engineers, software developers, and clinical researchers to bridge mechanical engineering principles with real-world surgical workflow requirements

Research Volunteer at PCIGITI

Sept. 2025 – May 2026

SickKids Hospital

Toronto, ON

- Assembled an ORCA open-source robotic hand, including 3D printing, wiring, and system calibration
- Documented build procedures, design changes, and test results to support future reproducibility and maintainability of the ORCA hand
- Recorded and presented project progress, identifying issues and proposing solutions

Project Manager, Engineering Design Team

Jan. 2025 – May 2025

Engineering Strategies and Practices II, University of Toronto

Toronto, ON

- Created and streamlined the project schedule and distributed tasks using Microsoft Project
- Spearheaded a team of six members to design and prototype a fully automated DSC crimper
- Scheduled team meetings, ensuring clear communication and alignment on project goals and deadlines
- Conducted comprehensive client meetings and site visits to collect vital data and understand client needs